Department of Information Systems
Bachelor of Information Systems
LAB3
IS301 Structured Programming In C

Weighting: 4%
ULO addressed: iii, iv vi, vii
Due date: 11:59 PM Thursday 17th April 2025
Total Marks: 20

Text Input/Output

Instructions:

1. This is your third lab activity.
2. Complete coding to solve the problem, and submit via Moodle before the due date.
3. You submit a single zipped file with the code in C and the input and output data files.
4. Based on the pseudocode develop an algorithm with the code.
5. Appendix A contains the pseudocode to guide you through the problem.
6. You can refer to Appendix B, a detailed breakdown of the marking criteria information.
7. By using both the rubric and the pseudocode, you should be well-equipped to successfully complete this lab activity.

Problem:

Using an editor, create the employee file with its headings as shown in the table below.

Employee No.	Department	Pay Rate	Exempt	Hours Worked
0101	41	8.11	Y	49
0722	32	7.22	N	40
1273	23	5.43	Y	39
2584	14	6.74	N	45

Write a C program to read employee data file and create a payroll register file. The register should contain the following data, with its text headings:

a) Employee number (print left-justified)
b) Department
c) Pay rate
d) Exempt
e) Hours Worked
f) Base Pay (pay rate * hours worked)

Appendix A: Pseudocode

```
BEGIN
   OPEN file "employee.txt" FOR reading AS inFile
   IF inFile is NULL THEN
      PRINT "Error opening employee.txt"
      TERMINATE program
   ENDIF

   OPEN file "payroll_register.txt" FOR writing AS outFile
   IF outFile is NULL THEN
      PRINT "Error creating payroll_register.txt"
      CLOSE inFile
      TERMINATE program
   ENDIF

   READ and discard the first line from inFile (headers)

   WRITE headers to outFile

   WHILE not end of inFile DO
      READ empNo, dept, payRate, exempt, hoursWorked
      CALCULATE basePay = payRate * hoursWorked
      WRITE empNo, dept, payRate, exempt, hoursWorked, basePay TO outFile
   ENDWHILE

   CLOSE inFile
   CLOSE outFile

   PRINT "Payroll register has been created."

END
```

Appendix B:

Marking Criteria

Criteria	Description	Marks
1. File Handling (Input)	Correctly opens the input file, reads content (including header handling).	2
2. File Handling (Output)	Correctly creates/writes to the output file with proper formatting.	3
3. Data Reading and Parsing	Reads each field accurately from the input file and stores values properly.	2
4. Base Pay Calculation	Correct calculation of base pay = pay rate × hours worked.	2
5. Output Formatting	Well-formatted output (columns aligned, headers included, readable layout).	3
6. Looping & Control Logic	Uses loop to iterate through all employees in the file.	2
7. Use of Variables & Data Types	Appropriate use of variable types for each data field.	1
8. Error Handling	Includes error handling for file opening/closing (e.g., file not found).	2
9. Code Structure and Comments	Organized code structure and relevant comments/documentation.	1
10. Compilation and Execution	Program compiles and runs correctly without errors.	1
11. Algorithm	A well-structured algorithm	1
TOTAL		**20**